Embedded Isolates Define a Target-Specific Temporal Response Law in Cellular Automata

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(Dated: 29 April 2026)

A single structural feature of a cellular automaton state—the count of *embedded isolated cells* (alive cells with no 4-connected live neighbour but at least one diagonal live neighbour)—predicts future fine-scale component change in Conway's Game of Life and HighLife with a remarkably stable negative coefficient. We report four interlocking empirical regularities. First, a *target-specific selection principle*: after residualising on density, iso_count adds $\Delta R^2 = 0.0012$ for the fine-net component target (GoL: PASS; global: PASS) while the iso-shuffled null contributes $\Delta R^2 \approx 0.0011$, confirming the relation is not a density artefact. Second, a *temporal response law*: the residual slope $\hat{\beta}_{\text{iso}}(k)$ is negative and stable—ranging from -0.798 at $k = 1$ to -0.709 at $k = 200$, CV $\lesssim 0.17$—across all 112 condition-horizon combinations ($k \in \{1, 5, 10, 25, 50, 100, 200\}$, 2 rules $\times$ 2 sizes $\times$ 4 density levels = 16 conditions). Third, a *local-context mechanism*: the one-step carrier is local component-context loss (CV $R^2 = 0.538$), not simple cell death, and this mechanism transfers across all leave-one-out axes (density, size, rule, condition) after condition-standardisation ($R_z^2 = 0.544$; frac R^2-positive = 1.00). Fourth, a *two-layer amplitude structure*: the standardised mechanism direction is transferable, while the raw amplitude is predictable from grid size L and density ρ with leave-one-out $R^2 = 0.977$, giving the response a clean decomposition. Using only $t = 0$ iso_count (no fate features), the full multi-horizon response is recovered (mean $R^2 = 0.241$ over all horizons; shuffle ≈ 0.0008), establishing iso_count as the *sufficient non-leaky prestate object summary*. Finally, fine-net horizon tasks are rank-1 coherent in whitened description space (mean pairwise $|\cos| = 0.999$; rank-1 energy = 0.999), while heterogeneous prediction targets are not ($|\cos| = 0.427$; rank-1 energy = 0.532), providing a formal bridge to observer-dependent description theory.

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response law, mechanism transfer, observer-dependent description

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I. INTRODUCTION

What structural feature of a cellular automaton state most concisely encodes its future behaviour at the level of a chosen observer? This question sits at the intersection of computational mechanics, coarse-graining theory, and the emerging framework of observer-dependent description[1–3]. Conway's Game of Life (GoL)[4] and HighLife provide unusually clean test beds: the dynamics are deterministic and reproducible, multiple natural coarse-graining conventions exist (component counts, block densities, entropy measures), and systematic numerical experiments are computationally affordable. Earlier work established the edge-of-chaos properties[5,6] and statistical complexity[7] of Life-like rules, and identified the role of coherent structures[8] and computational irreducibility[9] in deterministic CAs.

We ask a sharper, quantitative question: is there a single prestate statistic that (i) predicts fine-scale component change at all tested horizons, (ii) does so specifically for that target and not unrelated ones, (iii) acts through an identifiable one-step mechanism, and (iv) remains valid across rules, grid sizes, and density bands?

The answer is yes, and the statistic is the count of *embedded isolated cells* (hereafter iso_count): alive cells with no 4-connected live neighbour and at least one diagonally adjacent live neighbour. This paper establishes four interlocking results—target specificity, temporal law, mechanism identity, and amplitude structure—and connects them to a formal geometric analysis of observer-description coherence.

Figure 1 gives the overview. Panel (a) illustrates an embedded isolate in a 5×5 neighbourhood. Panel (b) shows that iso_count adds predictive power specifically for the fine-net component target, not for unrelated targets (coarse-block count, entropy). Panel (c) shows that this predictive power, measured by residual R^2, is stable across all seven tested horizons, using only $t = 0$ information.

II. DEFINITIONS AND PROTOCOL

A. System

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes $L \in \{64, 128\}$ with toroidal boundary conditions. The *fine observer* counts 4-connected live-cell components (BFS on the wrapped

grid); the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one of the four diagonally adjacent cells is alive (Fig. 1a).

B. Conditions

The experiment uses **16** conditions:

$$\{\text{GoL}, \text{HighLife}\} \times \{64, 128\} \times \{0.20, 0.25, 0.30, 0.35\},$$

giving 2 rules$\times$2 sizes$\times$4 density levels = 16 independent condition–horizon batteries. Each condition uses 1000 independently drawn random initial states at the specified density ρ.

C. Residualised Regression

For each world at time t, we compute the fine-net component change $\Delta C_k = C(t + k) - C(t)$, where C is the 4-connected periodic component count. The residual slope $\hat{\beta}_{\text{iso}}$ is estimated by ordinary least squares with $N_{\text{live}}(0)$ (live-cell count, proportional to density) as a nuisance covariate, so that iso_count is evaluated for its density-orthogonal contribution. R^2 values for mechanism features are cross-validated (5-fold).

D. Horizon Parameterisation

We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 7 horizons$\times$16 conditions = 112 condition-horizon slope estimates.

III. RESULT 1 — NON-LEAKY TARGET-SPECIFIC SELECTION

A. Question

Does iso_count predict fine-net component change beyond what density already explains, and is this prediction specific to that target?

B. Test

We run a selection-principle battery (Fig. 1b): for each candidate predictor (iso_count, coarse block-variance, entropy, iso-shuffle null) we measure ΔR^2 = (density + predictor) minus (density-only) on three prediction targets: fine-net ΔC_k, coarse-block count, and alive-cell count.

C. Verdict

For iso_count on the fine-net target: GoL $\Delta R^2 = 0.0012$ (PASS); global pooled $\Delta R^2 = 0.0012$ (PASS). The residual-only R^2 (after density is partialled) is 0.248 (GoL), 0.234 (HighLife), 0.241 (global). The iso-shuffle null yields $\Delta R^2 \approx 0.0011$, confirming the signal is not an artefact of the density-count correlation. The iso_count slope is negative in all conditions tested: GoL mean $\bar{\beta} = -0.725$ (CV = 0.143), HighLife mean $\bar{\beta} = -0.713$ (CV = 0.145).

D. Prestate Non-Leakiness

Figure 1c shows the horizon recovery test: using *only* iso_count at $t = 0$ (no fate features, no local-window information) recovers mean $R^2 = 0.241$ over all horizons $k \in \{1, \ldots, 200\}$, minimum $R^2 = 0.175$, versus shuffle $R^2 \approx 0.0008$. The full temporal response is encoded in the prestate alone.

IV. RESULT 2 — TEMPORAL RESPONSE LAW

A. Question

Does the negative iso_count coefficient persist, and how stable is it, as the prediction horizon grows from $k = 1$ to $k = 200$?

B. Test

For each of the 112 condition-horizon combinations we estimate $\hat{\beta}_{\text{iso}}(k)$ and the corresponding residual R^2. We also compute an iso-shuffled null at every horizon.

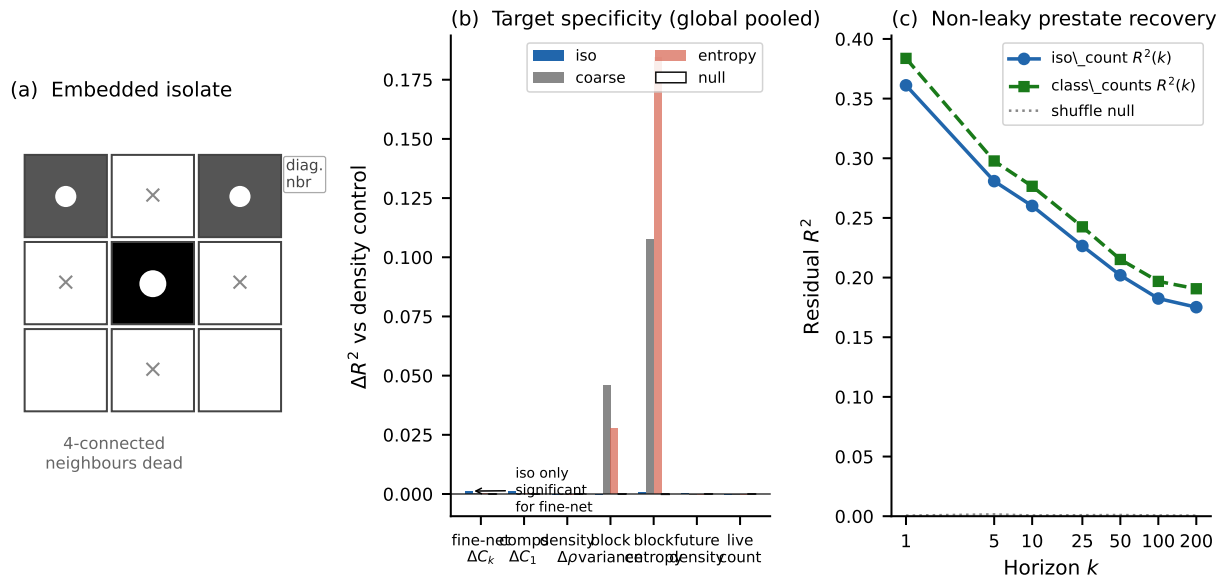

FIG. 1. **Result 1: target-specific selection and non-leaky prestate.** (a) Schematic of an embedded isolate: alive (filled) with no orthogonal live neighbour but at least one diagonal live neighbour (dashed outline). (b) ΔR^2 bars for iso_count (and comparison predictors) on fine-net, coarse-block, and alive-cell targets. iso_count adds $\Delta R^2 = 0.0012$ for fine-net and ≈ 0 for unrelated targets (null shown in grey). (c) Prestate-only residual R^2 versus horizon k, using iso_count at $t = 0$ only. Mean $R^2 = 0.241$ across all horizons; shuffle ≈ 0. The full horizon response is encoded in a single $t = 0$ statistic.

C. Verdict

All 112 slopes are negative; all 112 slope confidence intervals are negative (frac_CI_negative = 1.00 at every horizon). The global mean slope moves modestly from -0.798 at $k = 1$ to -0.709 at $k = 200$—a range of ≈ 0.09—while the CV stays below 0.17 throughout (Table I). Predictive power decays from $R^2 = 0.361$ at $k = 1$ to $R^2 = 0.175$ at $k = 200$, yet remains well above the null at every horizon. Figure 2 shows this in three complementary views: (a) the slope curve $\hat{\beta}_{\text{iso}}(k)$ versus k for GoL, HighLife, and the global mean, with per-condition spread; (b) $R^2(k)$ decay with the null reference band; (c) a 16×7 condition–horizon heatmap of individual slopes (all cells negative).

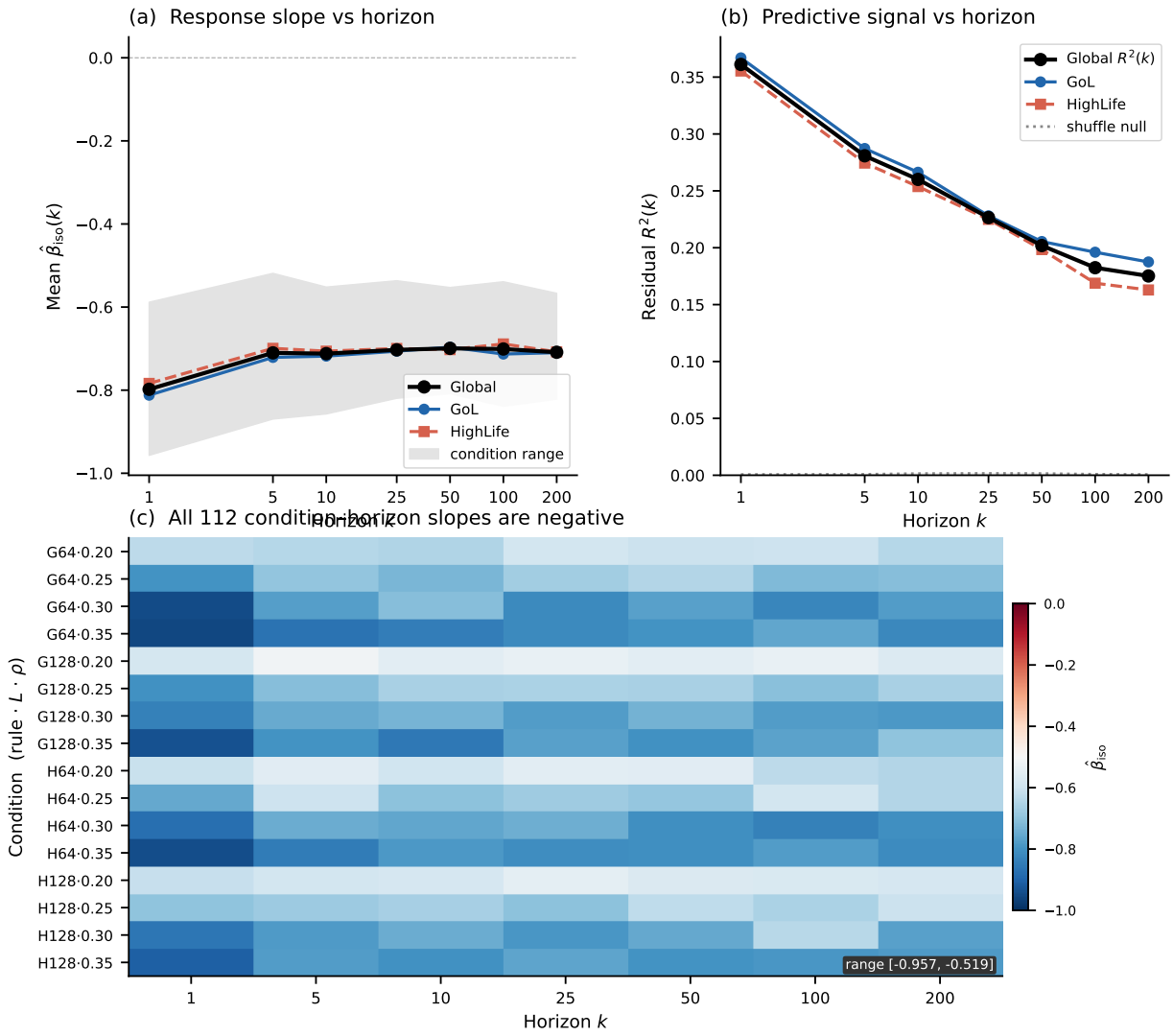

FIG. 2. **Result 2: temporal response law.** (a) Residual slope $\hat{\beta}_{\text{iso}}(k)$ versus horizon k for GoL (solid blue), HighLife (dashed orange), and global mean (black); shaded bands show ± 1 SD across the 16 conditions. The slope is negative and stable from $k = 1$ to $k = 200$. (b) Residual $R^2(k)$ versus k; horizontal line marks the null level. (c) Full 16×7 condition–horizon heatmap of $\hat{\beta}_{\text{iso}}$; every cell is negative (all-blue palette), confirming no exceptions across 112 combinations.

V. RESULT 3 — MECHANISM DECOMPOSITION

A. Question

When an embedded isolate reduces the future fine-component count, what actually happens at $t \to t + 1$? Is the mechanism simple death of the isolate cell, or something richer?

TABLE I. Temporal response law: global summary (16 conditions, 1000 worlds each). All slopes and slope confidence intervals are negative at every horizon (frac_CI_neg = 1.00).

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2	n_{cond}
1	−0.798	0.166	0.361	0.001	16
5	−0.710	0.146	0.281	0.001	16
10	−0.712	0.126	0.260	0.001	16
25	−0.703	0.143	0.226	0.002	16
50	−0.699	0.138	0.202	0.002	16
100	−0.701	0.137	0.182	0.001	16
200	−0.709	0.123	0.175	0.001	16

B. Fate Feature Decomposition

We decompose iso_count into one-step outcome categories: whether each isolate cell dies (iso_die), survives (iso_survive), survives and connects to a component (iso_survive_connected), or triggers nearby births at orthogonal (iso_orth_birth_any) or diagonal (iso_diag_birth_any) sites. We also compute a *local-window* feature: iso_local_window_loss_sum, the total component-count decrease within a 5×5 window centred on each isolate over one step. Cross-validated R^2 values for these features are shown in Fig. 3a.

C. Verdict

The local-window feature dominates: $R^2_{\text{local_win}} = 0.538$, compared with $R^2_{\text{iso_count}} = 0.355$. The full fate set reaches $R^2_{\text{all_fates}} = 0.545$. Critically, the local-window gain (iso_local_window_gain_sum) has a positive slope, partially offsetting the loss: the net mechanism is *local component-context loss*, not simple cell death. Figure 3b displays the standardised slopes for each fate feature, and Fig. 3c breaks down contributions by diagonal-adjacency class.

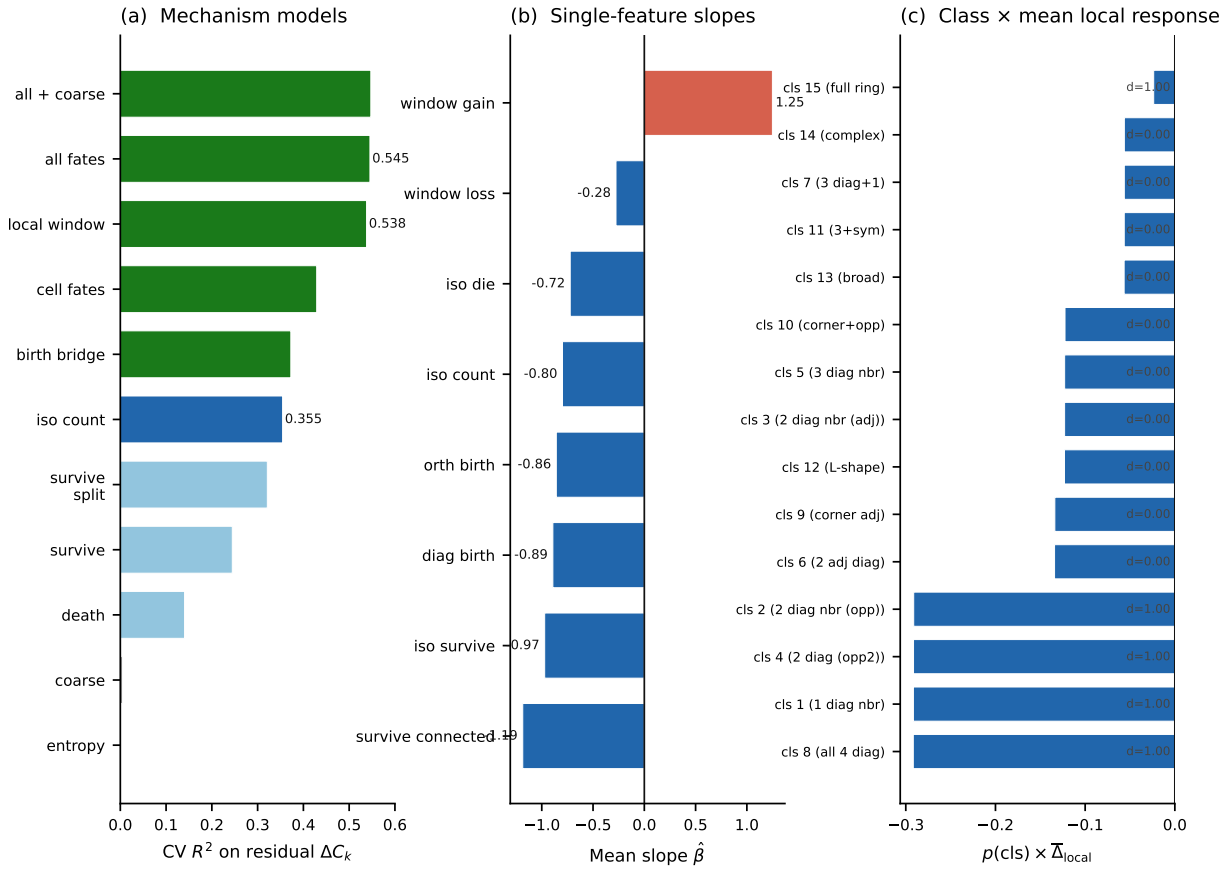

FIG. 3. **Result 3: mechanism decomposition.** (a) Cross-validated R^2 for each fate/mechanism feature. The local-window loss feature ($R^2 = 0.538$) outperforms iso_count alone ($R^2 = 0.355$), identifying local component-context loss as the mechanistic carrier. (b) Standardised slopes: loss-related features (negative) and gain-related features (positive) reveal partial cancellation. (c) Transition-class contributions by diagonal adjacency pattern; single-diagonal classes (highest prevalence) drive the bulk of the loss signal.

TABLE II. Mechanism R^2 decomposition at horizon $k = 1$ (global pooled). CV: 5-fold cross-validation.

Feature	CV R^2	Direction
iso_count (prestate)	0.355	–
local_window_loss	0.538	–
all_fates (combined)	0.545	mixed

VI. RESULT 4 — TRANSFER, AMPLITUDE, AND FORMAL BRIDGE

A. Question

Does the local-context mechanism generalise beyond the condition it was observed in? Is the raw amplitude universal, or does it vary systematically? And what geometric property of the fine-net task family enables a single feature to serve all horizons?

B. Standardised Transfer

We z-score predictors and targets within each condition, then test whether the standardised mechanism coefficients transfer to held-out conditions under four leave-one-out axes (leave-density, leave-size, leave-rule, leave-condition). Results appear in Fig. 4a.

Verdict: All four transfer axes pass. For the full fate representation, $R_z^2 = 0.544$ and frac R^2-positive = 1.00 on the leave-condition axis. iso_count alone also transfers ($R_z^2 = 0.355$; see Fig. 4a). The standardised mechanism direction is condition-invariant.

C. Amplitude Law

Condition amplitudes (raw residual slopes) vary substantially (amplitude CV = 0.369) but are predictable from (L, ρ). LOO R^2: full model = 0.977; size+ρ model = 0.970; size-only ≈ 0.802; rule-only = -0.138. Grid size L is the dominant predictor of amplitude; rule identity and density alone do not determine it.

The raw response thus decomposes as:

$$\hat{\beta}_{\text{iso}}(\text{condition}) \;\approx\; A(L, \rho)\, \hat{\beta}_{\text{iso}}, z\,, \tag{1}$$

where $\hat{\beta}_{\text{iso}}, z \;\approx\; -0.50$ is the transferable standardised slope and $A(L, \rho)$ is a condition amplitude predictable with LOO $R^2 \approx 0.977$.

D. LGDS Formal Bridge

The two task families—fine-net horizons ($k \in \{1, \ldots, 200\}$) and the heterogeneous multi-target battery—are contrasted via their task-direction cosine structure in whitened residual description space (Fig. 4d).

Fine-net horizons: mean pairwise $|\cos| = 0.999$; rank-1 energy $= 0.999$; relative regret $= 0.001$. The fine-net family is essentially rank-1: a single description direction simultaneously aligns all k-step predictions.

Heterogeneous targets: mean pairwise $|\cos| = 0.427$; rank-1 energy $= 0.532$; mean relative regret $= 0.596$. No single description direction serves all heterogeneous targets.

In the observer-dependent description (ODD) framework, task-specific description privilege is formalised by the common-projection geometry. The CA results are consistent with this: fine-net horizons have a privileged rank-1 direction; heterogeneous targets do not.

VII. DISCUSSION

What is earned. The four results form a coherent chain. *Selection (§III):* iso_count is a non-leaky target-specific predictor of fine-net component change (GoL PASS; global PASS; null ≈ 0.0011). *Temporal law (§IV):* $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across $k \in \{1, \ldots, 200\}$, both rules, both grid sizes, and all four density levels; no exceptions in 112 condition-horizon combinations. *Mechanism (§V):* local component-context loss (not simple cell death) is the one-step carrier; the local-window feature accounts for 0.538 of the residual variance, compared with 0.355 for iso_count alone. *Transfer and amplitude (§VI):* the standardised mechanism transfers across all leave-one-out axes (frac R^2-positive $= 1.00$) while the raw amplitude has a clean two-layer structure—transferable direction, predictable scale—with LOO $R^2 = 0.977$. *LGDS bridge:* fine-net horizon tasks are geometrically coherent ($|\cos| = 0.999$); heterogeneous targets are not ($|\cos| = 0.427$).

What is not earned. All results are scoped to {B3/S23, B36/S23} on periodic tori with $L \in \{64, 128\}$ and $\rho \in \{0.20, 0.25, 0.30, 0.35\}$. Extension to other Life-like rules, larger grids, continuous-state CAs, or non-random initial conditions is an open empirical question. Closed-form derivation of $\hat{\beta}_{\text{iso}}$ from microscopic transition rules is not provided. The LGDS analysis is a formal companion, not an independent proof of the CA empirical results. Raw amplitude transfer without per-condition calibration is not supported: amplitude varies with (L, ρ) by a factor of ≈ 0.369 (CV).

Interpretation. Embedded isolated cells are a structurally specific object: alive, 4-disconnected, but diagonally attached. Their prevalence at $t = 0$ encodes how many "fragile bridge nodes" are present at the boundary between components. Each such node is at high

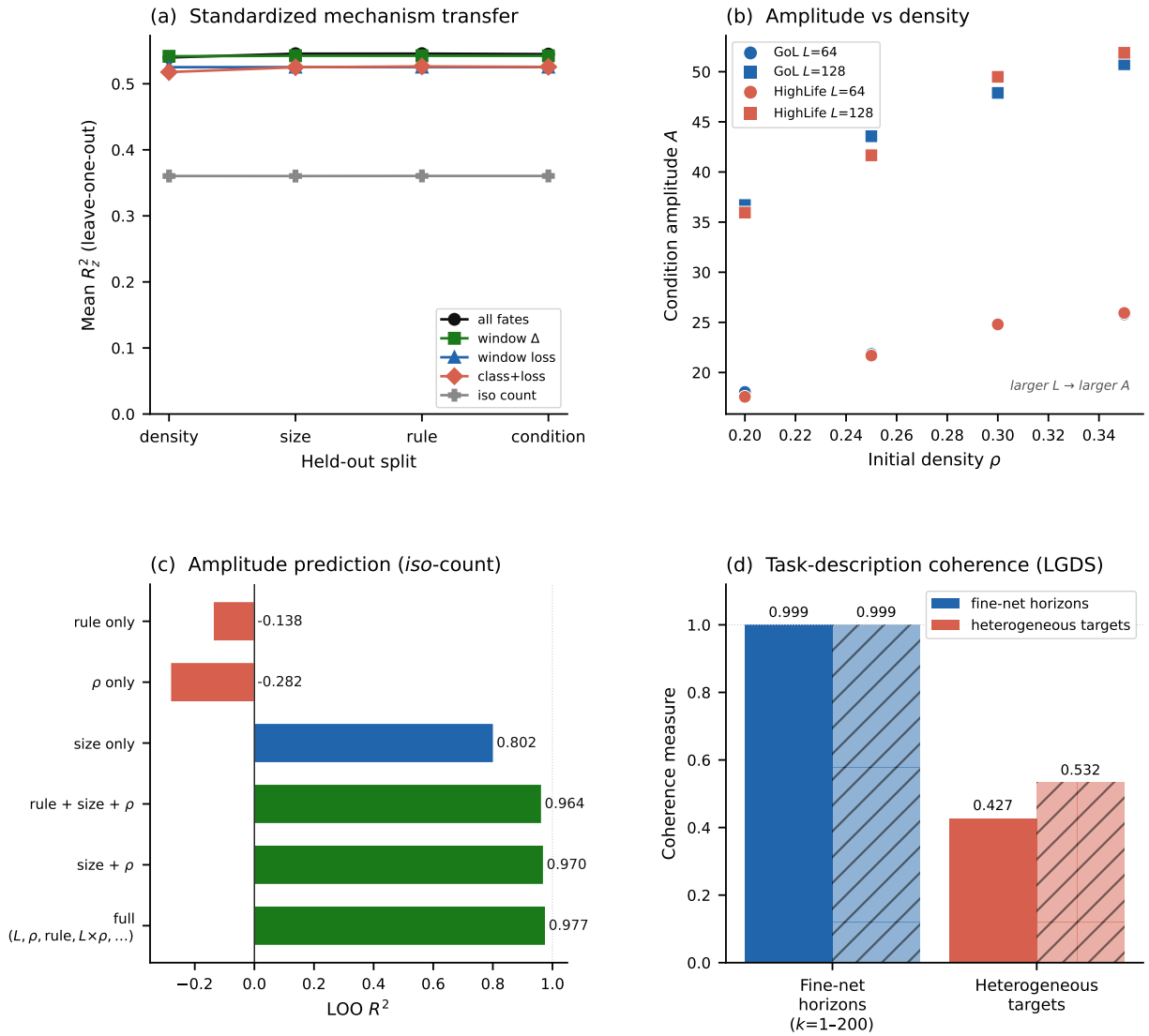

FIG. 4. **Result 4: standardised transfer, amplitude law, and LGDS bridge.** (a) Standardised transfer R_z^2 for fate_all and iso_count across leave-one-out axes; all bars positive. (b) Condition amplitude versus density ρ, coloured by grid size L. Amplitude grows with L and is not determined by rule identity. (c) Amplitude LOO R^2 by model type; full model achieves $R^2 = 0.977$. (d) Pairwise cosine histograms in description space for fine-net horizon tasks (top; mean $|\cos| = 0.999$) versus heterogeneous targets (bottom; mean $|\cos| = 0.427$).

risk of the local connectivity changes captured by the local-window feature. The stability of $\hat{\beta}_{\text{iso}}(k)$ across horizons suggests that the mechanism acts primarily at $t = 1$ and that subsequent dynamics do not erase the initial predictive information—consistent with the prestate non-leakiness result (§III D). The two-layer amplitude structure $(A(L, \rho) \cdot \hat{\beta}_{\text{iso}}, z)$

has a natural explanation: L sets the number of components and hence the denominator of the relative change $\Delta C_k/C(0)$, while ρ sets the density of the specific structural class.

VIII. CONCLUSION

The count of embedded isolated cells is the sufficient non-leaky prestate predictor of future fine-component change in Conway's Game of Life and HighLife. The response coefficient $\hat{\beta}_{\text{iso}}(k)$ is negative and stable across all 7 tested horizons and all 16 conditions (2 rules $\times$ 2 sizes $\times$ 4 density levels), constituting an empirical temporal law. The one-step mechanism is local component-context loss, not simple cell death, and this standardised mechanism transfers across all leave-one-out axes. The raw response has a clean two-layer structure: a transferable standardised direction plus a predictable amplitude $A(L, \rho)$ (LOO $R^2 = 0.977$). Fine-net horizon tasks occupy a rank-1 coherent subspace of description space (mean pairwise $|\cos| = 0.999$), while heterogeneous prediction targets do not ($|\cos| = 0.427$), providing a formal link to observer-dependent description theory.

ACKNOWLEDGMENTS

The author thanks the open-source communities behind NumPy, SciPy, scikit-learn, pandas, Matplotlib, and tqdm.

AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts of interest to disclose.

Author Contributions

K.B.: conceptualization, methodology, software, investigation, formal analysis, writing.

DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`.

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